Dynamic Clustering

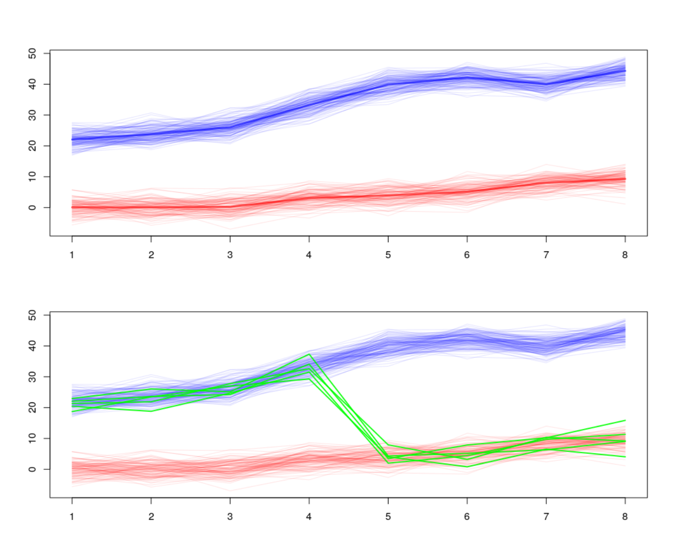
DATA 573

Longitudinal data

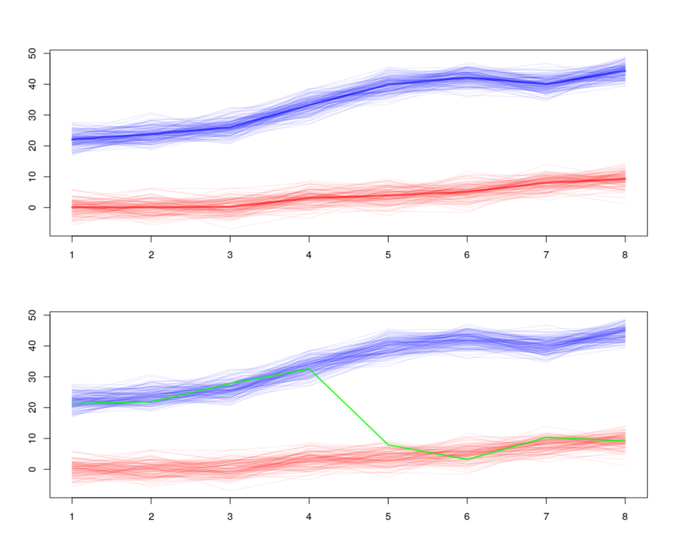
- Longitudinal studies record data over time
- Standard mixture models (that we've already learned) can be used on such data, but don't necessarily take advantage of known structures within this type of data.
- Some decompositions and constraints can have auto-regressive interpretations[1]
- What happens if we have observations that change groups over time?

[1] McNicholas, P.D. and Murphy, T.B., 2010. Model-based clustering of longitudinal data. *Canadian Journal of Statistics*, 38(1), pp.153-168.

Illustration



Illustration



Longitudinal Mixture Models

- Longitudinal mixture models generally fall into three categories[2]
- 'Mixture growth' — univariate densities for each timepoint that depend on the cluster
- 'Mixture Markov' — univariate densities conditional on both cluster and measurements from previous timepoints
- 'Hidden Markov' — observations are permitted to change clusters across timepoints, univariate densities are conditioned only on the cluster.
- The latter model is useful for dynamic clustering—unsupervised learning where observations are permitted to change groups over time.

[2]Vermunt, J. K. (2010). Longitudinal research using mixture models. In *Longitudinal research with latent variables*, pages 119–152. Springer.

A Simple HMM

- For observed $\mathbf{x}$ across T timepoints, a simple Gaussian-based HMM can be defined by

$$f(\mathbf{x}_i) = \sum_{g_0=1}^{G} \sum_{g_1=1}^{G} \cdots \sum_{g_T=1}^{G} P(z_{i0} = g_0) \left[\prod_{t=1}^{T} P(z_{it} = g_t \mid z_{i(t-1)} = g_{t-1}) \right] \left[\prod_{t=0}^{T} \phi(x_{it} \mid z_{it} = g_t) \right]$$

- $P(z_{i0} = g_0)$ is basically the same as prior probabilities (ratio of observations beginning in each state/cluster)
- $P(z_{it} = g_t \mid z_{i(t-1)} = g_{t-1})$ are ‘state/cluster transition probabilities’—if I’m in group A at the previous timepoint, what’s my probability that I transfer to group B at the next timepoint.
- $\phi(x_{it} \mid z_{it} = g_t)$ are just univariate Gaussians with parameters dependent on the state/group.

Hidden Markov Simulations

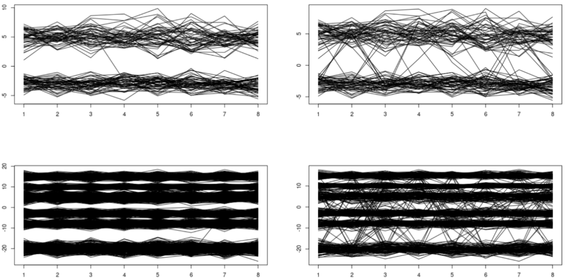


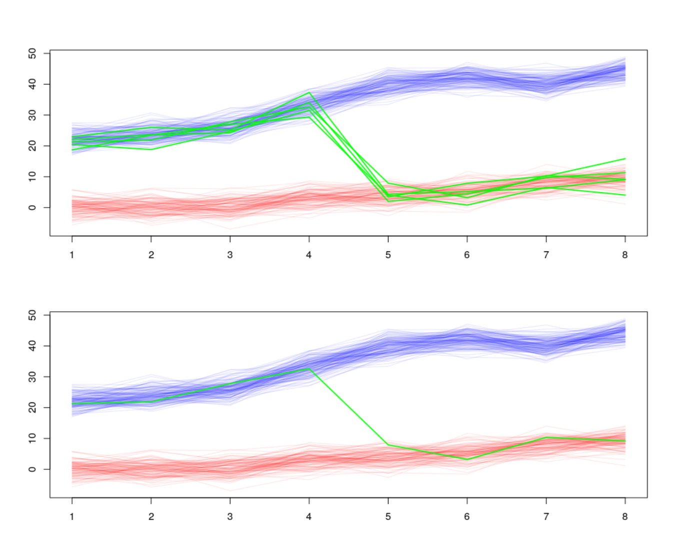
Figure 7: Examples for each of the hidden Markov model simulation scenarios. Clockwise from top left: 'Low Group, Unlikely Switch', 'Low Group, More Common Switch', 'Mid Group, More Common Switch', 'Mid Group, Unlikely Switch'.

Finite Gaussian Mixture Models

- The number of free parameters (especially in $\boldsymbol{\Sigma}_g$) is the primary reason why standard mixture models are unable to model dynamic observation behaviour.
- Let's introduce a clustering framework that allows (even small numbers of) observations to change groups over time, but retains ML estimation of covariances (and other parameters) within a (mostly) standard EM algorithm.[3]
- We are going to *abuse* the idea of missing data in order to get there...

[3] Welsh, Andrews, and Browne. Modelling group-switching in cluster analysis via compositional mixture components. To appear in *Journal of Classification*.

Illustration, Again



Summary of Modelling Approach

- The plan is...
  1. Perform standard Gaussian model-based clustering
  2. Seek and flag observations that have (potentially) changed groups over time
  3. Propose and re-fit with new mixture components that share parameters with the original components.
- Steps 1 and 2 are fairly straightforward
- For Step 3, we will reformulate the model with a twisted take on 'missing data' to achieve our goals.

Summary of Modelling Approach

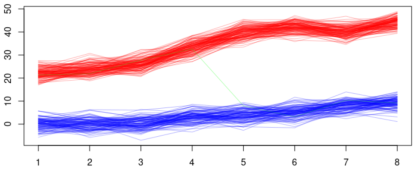


Figure 2: A simple motivating example to illustrate a single observation exhibiting group-switching behaviour.

	V1	V2	V3	V4	V5	V6	V7	V8
Green Observation	21.33	21.89	27.79	32.62	7.92	3.19	10.35	9.19
Green Obs. Part 1	21.33	21.89	27.79	32.62	NA	NA	NA	NA
Green Obs. Part 2	NA	NA	NA	NA	7.92	3.19	10.35	9.19

Table 1: The first row shows the green observation in 'concatenated' form. The second and third row provide the observation in its 'split' form.

Step 1: Cluster

- Initially, we cluster with a traditional p-dimensional multivariate Gaussian mixture model.
- That is, we assume

$$f(\mathbf{x}) = \sum_{g=1}^{G} \tau_g \phi_p(\mathbf{x} \mid \boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g),$$

- As usual, a cluster membership indicator variable is introduced and considered to be missing data. $Z_{ig} = 1$ if observation i belongs to group g, 0 otherwise.

Step 1: Cluster - EM

- E-step:

$$E[Z_{ig}|\mathbf{x}_i, \boldsymbol{\theta}_g] = \frac{\hat{\tau}_g \phi_p(\mathbf{x}_i|\hat{\boldsymbol{\mu}}_g, \hat{\boldsymbol{\Sigma}}_g)}{\sum_{g=1}^{G} \hat{\tau}_g \phi_p(\mathbf{x}_i|\hat{\boldsymbol{\mu}}_g, \hat{\boldsymbol{\Sigma}}_g)} =: \hat{z}_{ig}$$

- M-step:

$$\hat{\tau}_g = \frac{\sum_{i=1}^{n} \hat{z}_{ig}}{n}$$

$$\hat{\boldsymbol{\mu}}_g = \frac{\sum_{i=1}^{n} \hat{z}_{ig}\mathbf{x}_i}{\sum_{i=1}^{n} \hat{z}_{ig}}$$

$$\hat{\boldsymbol{\Sigma}}_g = \frac{\sum_{i=1}^{n} \hat{z}_{ig}(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_g)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_g)'}{\sum_{i=1}^{n} \hat{z}_{ig}}$$

Brief Comments

- With sufficient observations changing groups in unison, Step 1 will ‘find’ a separate component necessary for them via model selection (such as BIC)...
- But what if this is rare, or more individualistic, behaviour?

Step 2: Seek and Flag

- With the mixture model arising from Step 1, we now look for observations that may be shifting among the groups.
- A naive search for this is to consider each time point in a marginal fashion, and compute the marginal version of the Z_{ig}. Introduce this as M_{igt}

$$\hat{m}_{igt} = \frac{\hat{\tau}_g \phi_1(x_{it}|\hat{\mu}_{gt}, \hat{\sigma}_{gt})}{\sum_{g=1}^{G} \hat{\tau}_g \phi_1(x_{it}|\hat{\mu}_{gt}, \hat{\sigma}_{gt})}$$

- This, admittedly, ignores correlation across variables, but is merely used to propose potential group-changing patterns...

Step 3: Propose and Re-fit

- Each potential group-changing pattern could then be introduced as a new component in the mixture model (we will discuss a stepwise procedure to explore this model space momentarily)
- Importantly, each new component is proposed only via sharing mean and covariance parameters with the original components.
- We are calling these new components ‘compositional components’.
- In order to fit within an EM framework, the notation will get nasty...
- Ultimately, we estimate ‘base’ component parameters using the ‘split-form’ observations with missing-at-random assumptions.

Step 3: Propose and Re-fit - Details

- Let H be the number of compositional components proposed for the new model, indexed by $h = 1, \ldots, H$.
- Let $\mathcal{S}_h = \{\mathcal{S}_{h1}, \ldots, \mathcal{S}_{hG}\}$ be a set of G subsets from set $\mathcal{V} = \{1, \ldots, p\}$ for each $h = 1, \ldots, H$.
- For each h we have $\mathcal{S}_{h1} \cup \ldots \cup \mathcal{S}_{hG} = \mathcal{V}$
- For each pair k & l such that $k \neq l$ we have that $\mathcal{S}_{hk} \cap \mathcal{S}_{hl} = \emptyset$
- Also, define $\mathcal{S}_{hg}^c$ the complement set of $\mathcal{S}_{hg}$ then $\mathcal{S}_{hg} \cup \mathcal{S}_{hg}^c = \mathcal{V}$.

Step 3: Propose and Re-fit - Details

- This allows us to describe parameters of the original mixture components via submatrices/vectors.
- For example, suppose that $\mathcal{S}_{hg}$ contains the first $d < p$ elements of $\{1, \ldots, p\}$

$$\boldsymbol{\Sigma}_g = \begin{bmatrix} \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg},\mathcal{S}_{hg})} & \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c,\mathcal{S}_{hg})} \\ \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg},\mathcal{S}_{hg}^c)} & \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c,\mathcal{S}_{hg}^c)} \end{bmatrix} \quad \text{and} \quad \boldsymbol{\mu}_g = \begin{bmatrix} \boldsymbol{\mu}_g^{(\mathcal{S}_{hg})} & \boldsymbol{\mu}_g^{(\mathcal{S}_{hg}^c)} \end{bmatrix}.$$

- Or, equivalently in this case...

$$\boldsymbol{\Sigma}_g = \begin{bmatrix} \boldsymbol{\Sigma}_g^{(1:d,1:d)} & \boldsymbol{\Sigma}_g^{(1:d,d+1:p)} \\ \boldsymbol{\Sigma}_g^{(d+1:p,1:d)} & \boldsymbol{\Sigma}_g^{(d+1:p,d+1:p)} \end{bmatrix} \quad \text{and} \quad \boldsymbol{\mu}_g = \begin{bmatrix} \boldsymbol{\mu}_g^{(1:d)} & \boldsymbol{\mu}_g^{(d+1:p)} \end{bmatrix}.$$

Step 3: Propose and Re-fit - Details

- Let the compositional components (formed to capture group-changing observations) be formulated via the product of marginals

$$g_p(\mathbf{x}) = \phi_{s_{h1}}(\mathbf{x}^{(\mathcal{S}_{h1})}|\boldsymbol{\mu}^{(\mathcal{S}_{h1})}, \boldsymbol{\Sigma}^{(\mathcal{S}_{h1})}) \dots \phi_{s_{hG}}(\mathbf{x}^{(\mathcal{S}_{hG})}|\boldsymbol{\mu}^{(\mathcal{S}_{hG})}, \boldsymbol{\Sigma}^{(\mathcal{S}_{hG})})$$

  such that $\sum_{g=1}^{G} s_{hg} = p$

- Resulting in the following mixture model...

$$f(\mathbf{x}) = \sum_{g=1}^{G} \tau_g \phi_p(\mathbf{x}|\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g) + \sum_{h=1}^{H} \tau_h \prod_{g=1}^{G} \phi_{s_{hg}}\left(\mathbf{x}^{(\mathcal{S}_{hg})}|\boldsymbol{\mu}_g^{(\mathcal{S}_{hg})}, \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg})}\right)$$

Two Huge Implications

1. Product of marginals assumption means that we assume independence for compositional components among the subsets of variables that are borrowed from differing base components.

- This is NOT the same as local independence from HMMs, where each timepoint is independent.

2. The compositional components only require H additional parameters to be estimated (the τ_h)!

Brief Contrast: HMMs vs Compositional

- τ_h are not quite the same as state-change probabilities.
- They provide the proportion of observations from the population that share the exact same state-change pattern.
- State-change probabilities from HMMs instead provide the probability of changing from one group to another at any time.

Brief Contrast: HMMs vs Compositional

- The number of free parameters for HMMs vs Compositional grow in different ways.
- HMMs: transition probabilities have $G^2 - G$ free parameters
- Compositional: compositional components have H free parameters
- Thus, if the number of group-switching patterns is low, the compositional approach has an advantage (especially as the number of groups increases).

Step 3: Propose and Re-fit - Details

- For $k = 1, \ldots, G, \ldots, G + H$, let

$$n_k = \sum_{i=1}^{n} \hat{z}_{ik}$$

$$\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{i=1}^{n} \hat{z}_{ik} \mathbf{x}_i$$

$$\mathbf{S}_k = \frac{1}{n_k} \sum_{i=1}^{n} \hat{z}_{ik} \left(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_k\right) \left(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_k\right)'$$

Step 3: Propose and Re-fit - Details

- E-step is unchanged, void of additional components...

$$\hat{z}_{ik} = \frac{\hat{\tau}_k \phi_p(\mathbf{x}_i | \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)}{\sum_{k=1}^{G+H} \hat{\tau}_k \phi_p(\mathbf{x}_i | \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)}$$

- M-step: $\boldsymbol{\mu}_k$

$$\widehat{\boldsymbol{\mu}}_g = \frac{n_g \overline{\mathbf{x}}_g + \sum_{h=1}^{H} n_{G+h} \widehat{\mathbf{w}}_{hg}}{n_g + \sum_{h=1}^{H} n_{G+h}}$$

where elements of the vector $\widehat{\mathbf{w}}_{hg}$ are defined as

$$\begin{aligned}
\widehat{\mathbf{w}}_{hg}^{(\mathcal{S}_{hg})} &= \overline{\mathbf{x}}_{G+h}^{(\mathcal{S}_{hg})} \\
\widehat{\mathbf{w}}_{hg}^{(\mathcal{S}_{hg}^c)} &= \boldsymbol{\mu}_g^{(\mathcal{S}_{hg}^c)} + \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})} \left[\boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg})} \right]^{-1} \left[\overline{\mathbf{x}}_{G+h}^{(\mathcal{S}_{hg})} - \boldsymbol{\mu}_g^{(\mathcal{S}_{hg})} \right].
\end{aligned}$$

Step 3: Propose and Re-fit - Details

- M-step: $\boldsymbol{\Sigma}_g$

$$\widehat{\boldsymbol{\Sigma}}_g = \frac{n_g \mathbf{S}_g + \sum_{h=1}^{H} n_{G+h} \widehat{\mathbf{T}}_{hg}}{n_g + \sum_{h=1}^{H} n_{G+h}}$$

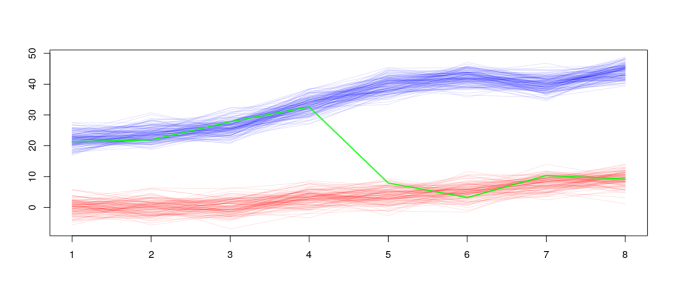
where the the elements of the matrix $\widehat{\mathbf{T}}_{hg}$ are defined as

$$\begin{aligned}
\widehat{\mathbf{T}}_{hg}^{(\mathcal{S}_{hg})} &= \widehat{\mathbf{S}}_{G+h}^{(\mathcal{S}_{hg})} \\
\widehat{\mathbf{T}}_{hg}^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})} &= \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})} \left[\boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg})}\right]^{-1} \widehat{\mathbf{S}}_{G+h}^{(\mathcal{S}_{hg})} \\
\widehat{\mathbf{T}}_{hg}^{(\mathcal{S}_{hg}, \mathcal{S}_{hg}^c)} &= \left[\widehat{\mathbf{T}}^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})}\right]' \\
\widehat{\mathbf{T}}^{(\mathcal{S}_{hg}^c)} &= \left(\boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})} \left[\boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg})}\right]^{-1} \widehat{\mathbf{S}}_{G+h}^{(\mathcal{S}_{hg})} - \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c, \mathcal{S}_{hg})}\right) \left[\boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg})}\right]^{-1} \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}, \mathcal{S}_{hg}^c)} + \boldsymbol{\Sigma}_g^{(\mathcal{S}_{hg}^c}
\end{aligned}$$

Model Search

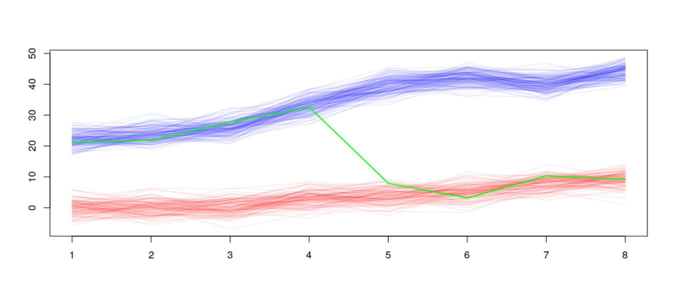
- Okay—one major problem is that Step 2 (which flags possible group-switching patterns) will, especially with noisy data, propose WAY too many patterns.
- We propose a mixed stepwise procedure.
  1. Add the compositional component with the largest observed likelihood
  2. Refit, compute BIC
  3. Until (current BIC) $<$ (best BIC - 15), return to 1
  4. Trim component with smallest τ_j.
  5. Refit, compute BIC
  6. If there are any empty components, return to 4
  7. Report model with best BIC

Motivating Simulation



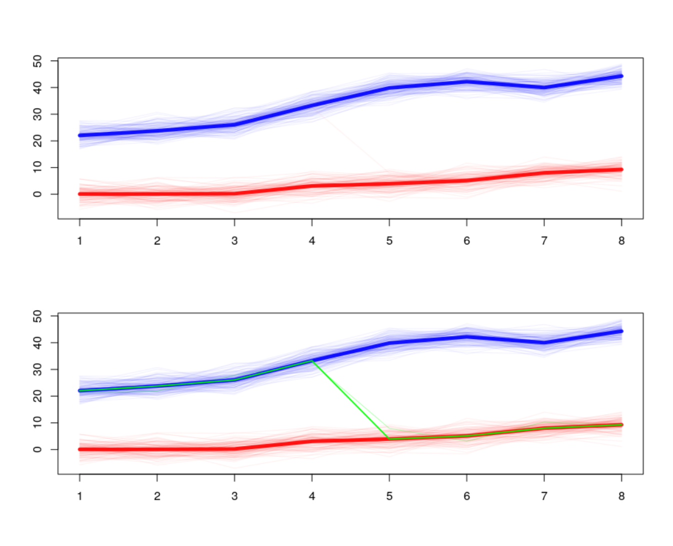
- We fit via the proposed 3-step process, using MCLUST with BIC as model-selection for Step 1 and reporting the BIC from our proposed procedure.

Motivating Simulation Results



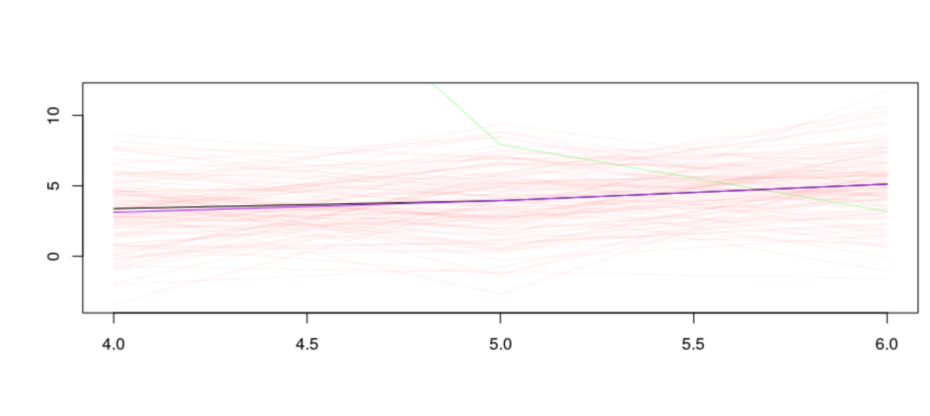
	MCLUST			Compositional		
	2	1		2	1	3
Blue	138	0		138	0	0
Red	0	111		0	111	0
Green	0	1		0	0	1
BIC	-9773.7			-9614.3		

Motivating Simulation Results



Motivating Simulation Means Zoom

- Black = MCLUST, Purple = Proposed



Motivating Simulation Covariances

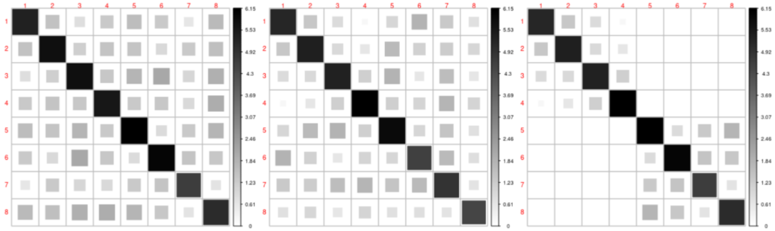
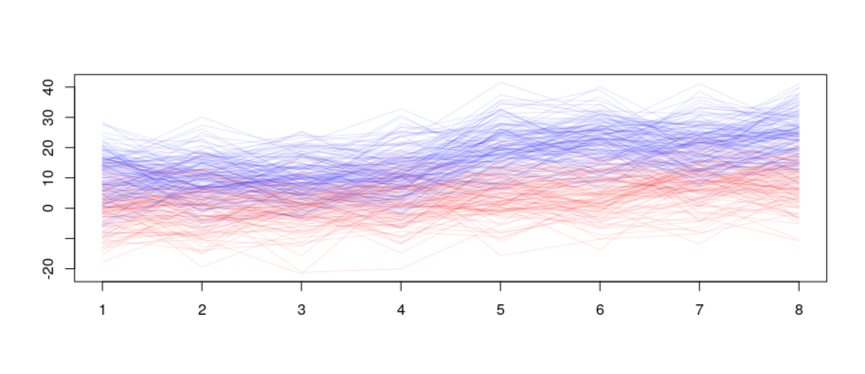


Figure 5: Visualization of the covariance matrices estimated using the compositional model. From left-to-right, they are estimated from the 'blue' base component, 'red' base component, and 'green' compositional component.

Noisy Simulation



Noisy Simulation Results

	MCLUST			Compositional	
	1	2		1	2
Blue	86	53		86	53
Red	1	110		1	110
BIC	-14,117.1			-14,117.1	

- Compositional model proposes 21 new clusters! But stepwise procedure (and BIC) recognizes their lack of utility.

Hidden Markov Simulations

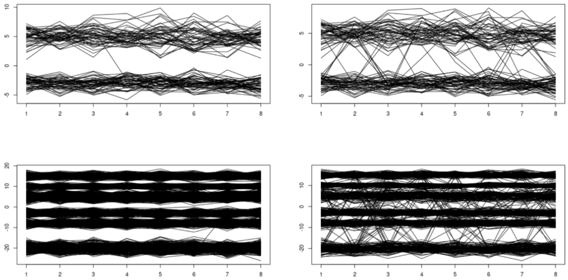


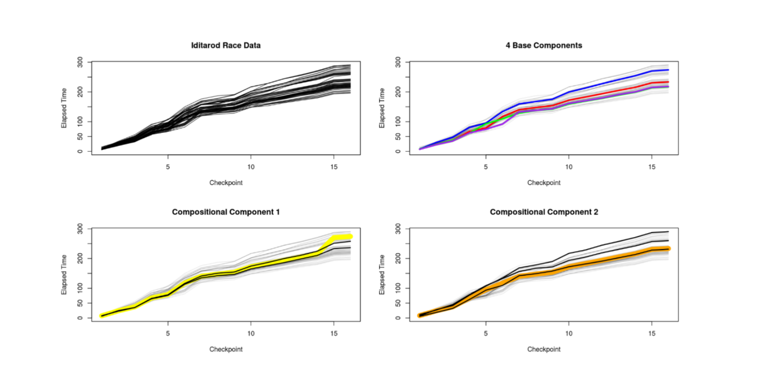
Figure 7: Examples for each of the hidden Markov model simulation scenarios. Clockwise from top left: 'Low Group, Unlikely Switch', 'Low Group, More Common Switch', 'Mid Group, More Common Switch', 'Mid Group, Unlikely Switch'.

Number of times Compositional approach wins (according to BIC, same order as figure, out of 100)

100	1
81	0

Real Data: Iditarod

Model	G	Decomp	BIC
MCLUST	2	VVE	-5183
longclust	2	EVA	-5156
Compositional	6	VE	-5120



Real Data: Bank

- Best compositional model beats MCLUST (BIC difference of 12.2). Second best compositional model (BIC difference of 2) shown below.

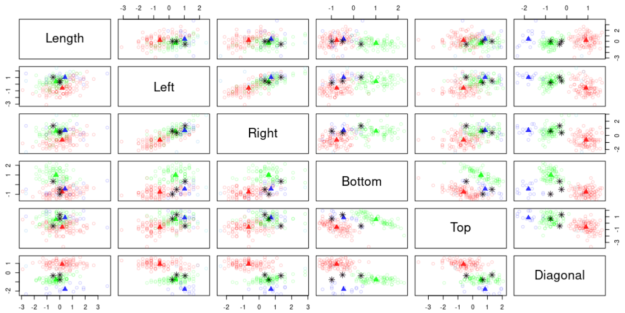


Figure 8: Scatterplot matrix of the Bank data set, highlighting the uniqueness of the observed pattern for observations 70, 103, and 125 (black asterisks). Base model component means are shown as filled triangles, along with observations assigned to those components in the same colour.

Real Data: Bank

- Viewed as outlier detection, the compositional approach beats contaminated mixture models and OCLUST in terms of ARI, and BIC (where applicable)

- This makes some sense because compositional components can detect a strange pattern on a subset of variables

- ContaminatedMixt works best if outliers are extreme across all variables for the static group that they are assigned to.

Real Data: Bank

- Note: lack of block-diagonal structure!

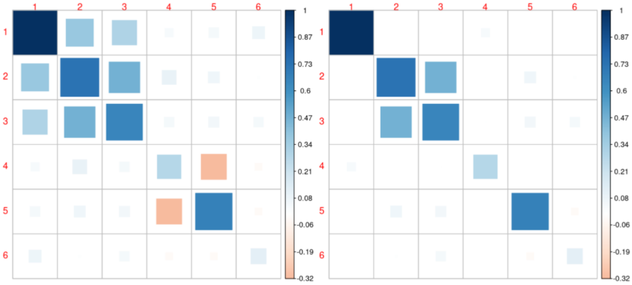


Figure 9: Visualization of the estimated covariance matrices for the scaled bank data set. Left is the estimated covariance for all base components under the UE model, right is the estimated covariance for compositional component 6 which contains observations 70, 103, and 125.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